Energy above hull and formation energy prediction analysis

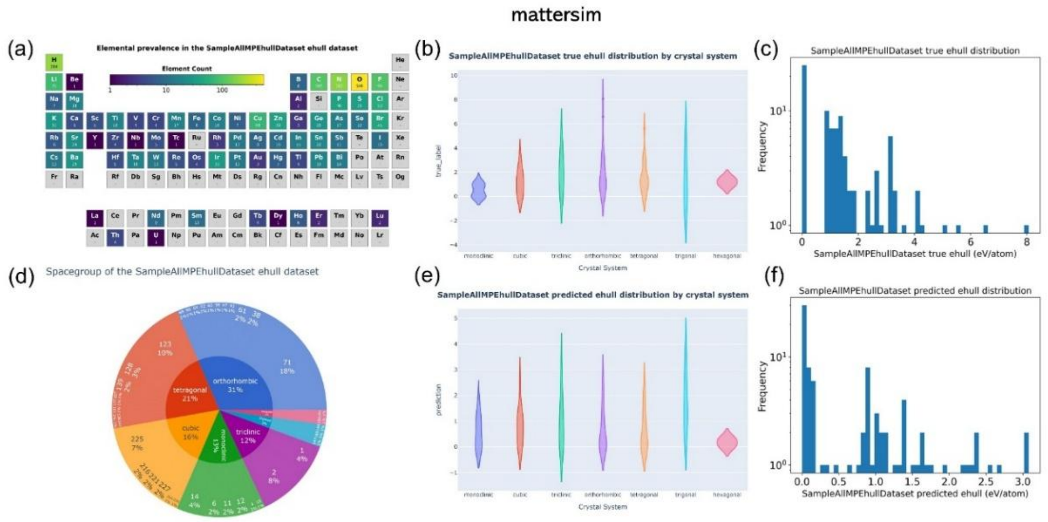

Figure S1 shows the visualization of the element coverage distribution of the 90 structures with prediction errors exceeding five times the average error in the MatterSim model in the general test dataset, as well as the pie chart of the energy above hull dataset based on crystal system and corresponding space group classification. Additionally, it includes a comparison of the distribution maps of the true and predicted energy above hull values based on crystal system classification, and a comparison of the frequency distribution histograms of the true and predicted energy above hull values.

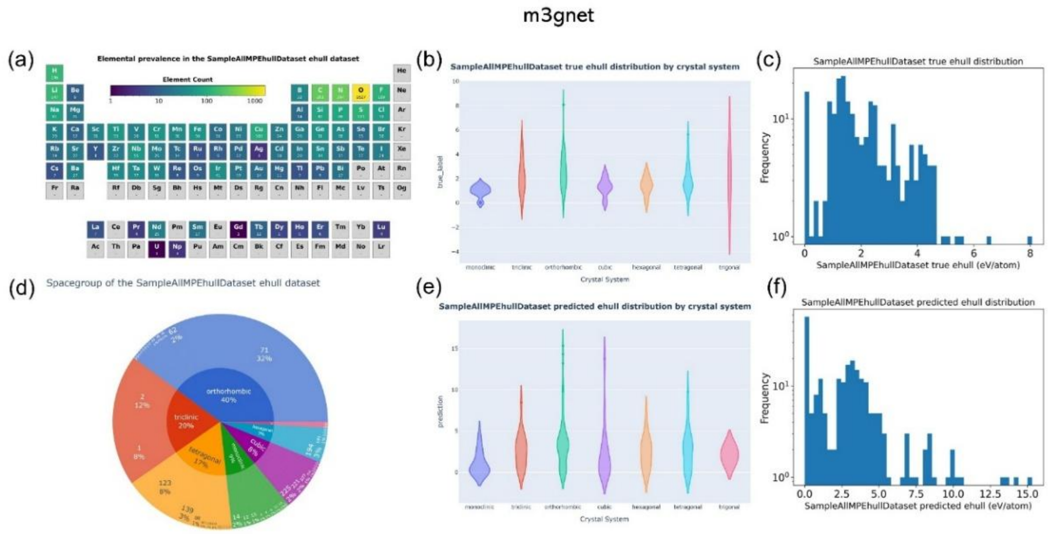

Figure S2 shows the visualization of the element coverage distribution of the 229 structures with prediction errors exceeding five times the average error in the m3gnet model in the general test dataset, as well as the pie chart of the energy above hull dataset based on crystal system and corresponding space group classification. Additionally, it includes a comparison of the distribution maps of the true and predicted energy above hull values based on crystal system classification, and a comparison of the frequency distribution histograms of the true and predicted energy above hull values.

mattersim

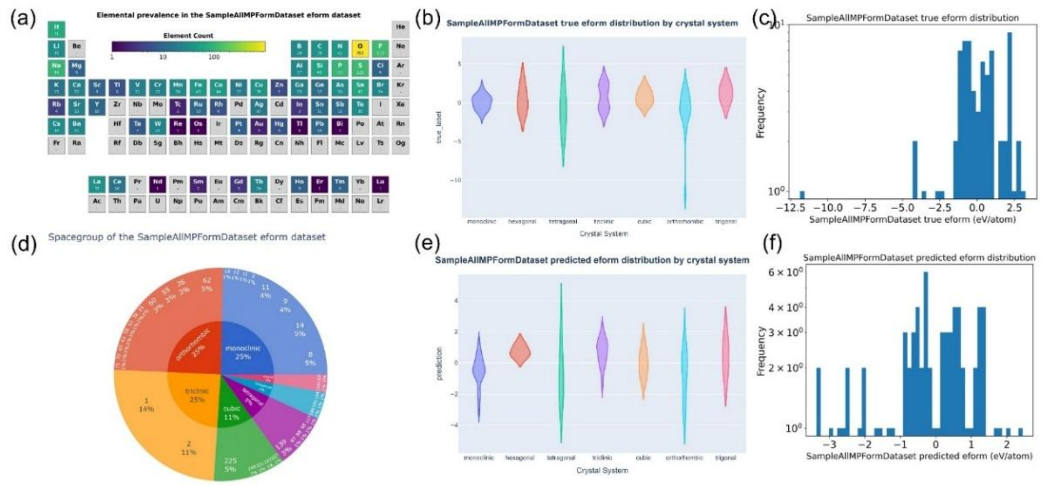

Figure S3 shows the visualization of the element coverage distribution of the 73 structures with prediction errors exceeding five times the average error in the MatterSim model in the general test dataset, as well as the pie chart of the formation energy dataset based on crystal system and corresponding space group classification. Additionally, it includes a comparison of the distribution maps of the true and predicted formation energy values based on crystal system classification, and a comparison of the frequency distribution histograms of the true and predicted formation energy values.

m3gnet

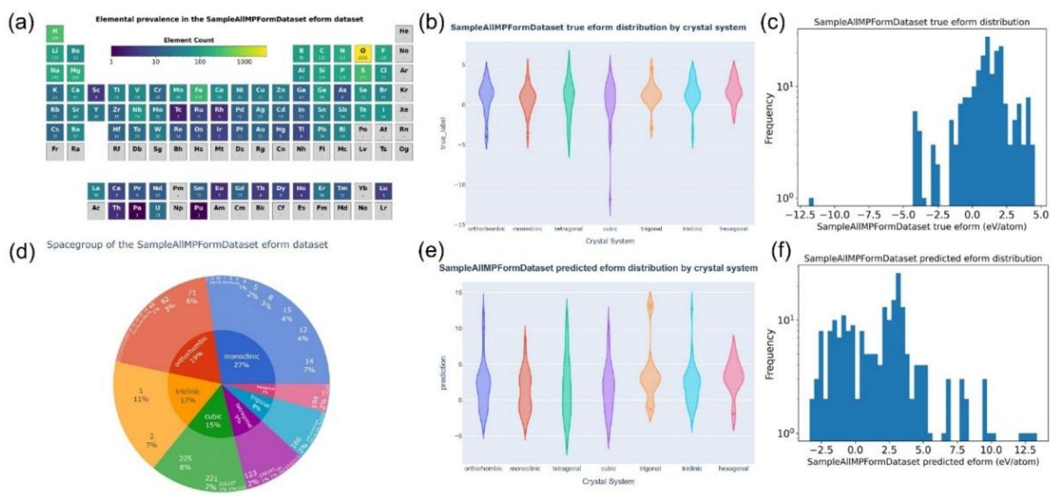

Figure S4 shows the visualization of the element coverage distribution of the 213 structures with prediction errors exceeding five times the average error in the SevenNet model in the general dataset, as well as the pie chart of the formation energy dataset based on crystal system and corresponding space group classification. Additionally, it includes a comparison of the distribution maps of the true and predicted formation energy values based on crystal system classification, and a comparison of the frequency distribution histograms of the true and predicted formation energy values.